M|P|W

Physics A

Ch.5 - Work, Energy and Power
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **35**.
- The total number of marks may take into account some 'either/or' question choices.

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1. A force of 12 N moves an object at an angle θ to the force. The object travels 9.6 m and the work done by the force is 52 J.

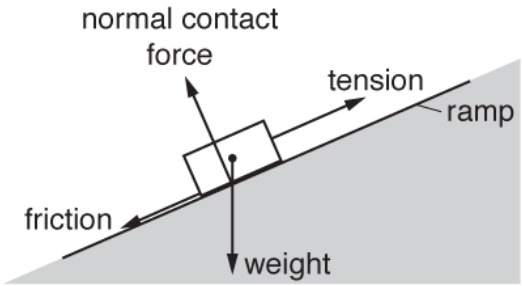
What is the angle θ ?

A 1.1°
B 27°
C 63°
D 90°

Your answer ☐ **[1]**

2. A block moves at **constant** speed up a ramp.

The diagram below shows all the forces acting on the block.



Which force does no work on, or against, the object as it travels up the ramp?

A weight
B friction
C tension
D normal contact force

Your answer ☐ **[1]**

3. The diagram below shows a rotating steam generator.

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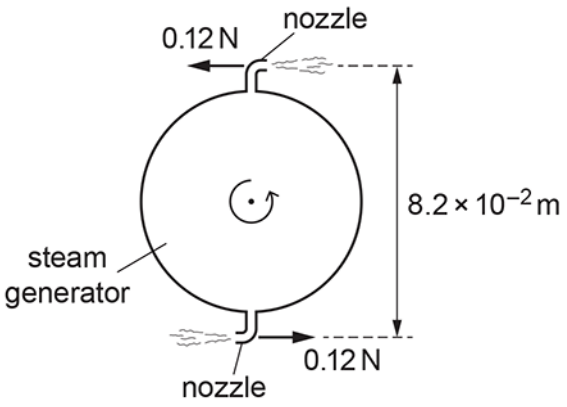


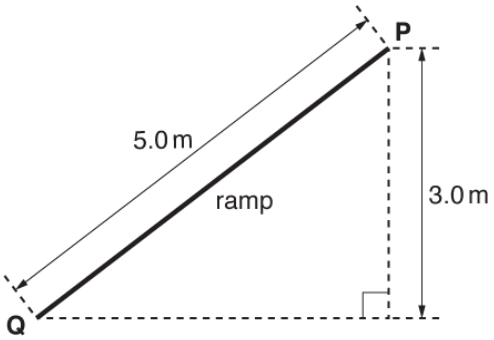
The steam ejected from the nozzles provides a couple. The force at each nozzle is 0.12 N. The perpendicular distance between the nozzles is 8.2×10^{-2} m.

What is the work done by the forces as the steam generator completes one revolution?

A 0 J
B 9.8×10^{-3} J
C 3.1×10^{-2} J
D 6.2×10^{-2} J

Your answer ☐ **[1]**

4. An object is at the top of a ramp at point **P**. The gravitational potential energy of the object at **P** is 100 J. The object is released from rest at **P**. It travels down the ramp. The kinetic energy of the object at the bottom of the ramp at point **Q** is 60 J.



What is the average resistive force acting on the object as it travels down the ramp?

A 8.0 N
B 10 N
C 12 N
D 20 N

Your answer ☐ **[1]**

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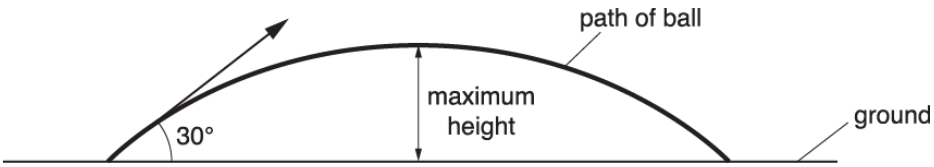
5. The frictional force acting on an object falling vertically through water is directly proportional to its speed squared.

What is the correct relationship between *P*, the rate of work done against the frictional force, and the speed *v* of the object?

A $P \propto v^{-1}$
B $P \propto v$
C $P \propto v^2$
D $P \propto v^3$

Your answer ☐ **[1]**

6. A ball is thrown at an angle of 30° to the horizontal. The initial kinetic energy of the ball is *K*. Air resistance has negligible effect on the motion of the ball.



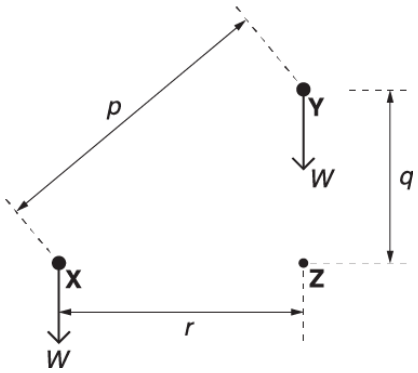
What is the kinetic energy of the ball at the maximum height?

A 0
B 0.25 *K*
C 0.75 *K*
D 0.87 *K*

Your answer ☐

[1]

7. A crane is used to lift a load directly from point **X** to point **Y**.



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The weight of the load is *W*.
p, *q* and *r* are distances between points **X**, **Y** and **Z** as shown in the diagram.

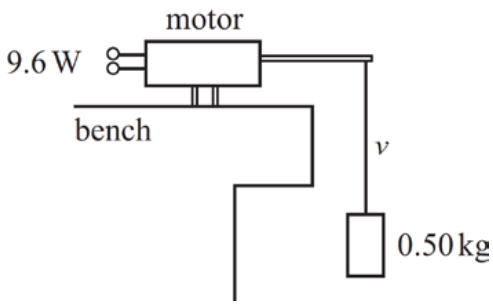
What is the work done against the weight?

A *Wp*
B *Wq*
C *Wr*
D *W(q + r)*

Your answer ☐

[1]

8. A small electric motor is 20% efficient. Its input power is 9.6 W when it is lifting a mass of 0.50 kg at a steady speed *v*.



What is the value of *v*?

A. 0.39 m s^{-1}
B. 2.0 m s^{-1}
C. 2.8 m s^{-1}
D. 3.8 m s^{-1}

Your answer ☐

[1]

9(a). A student investigates the motion of a tennis ball of mass 57 g which falls vertically from rest, then bounces once on a soft horizontal surface.
Fig. 1 shows the variation with time *t* of the velocity *v* of the tennis ball falling from rest until it hits the soft surface.

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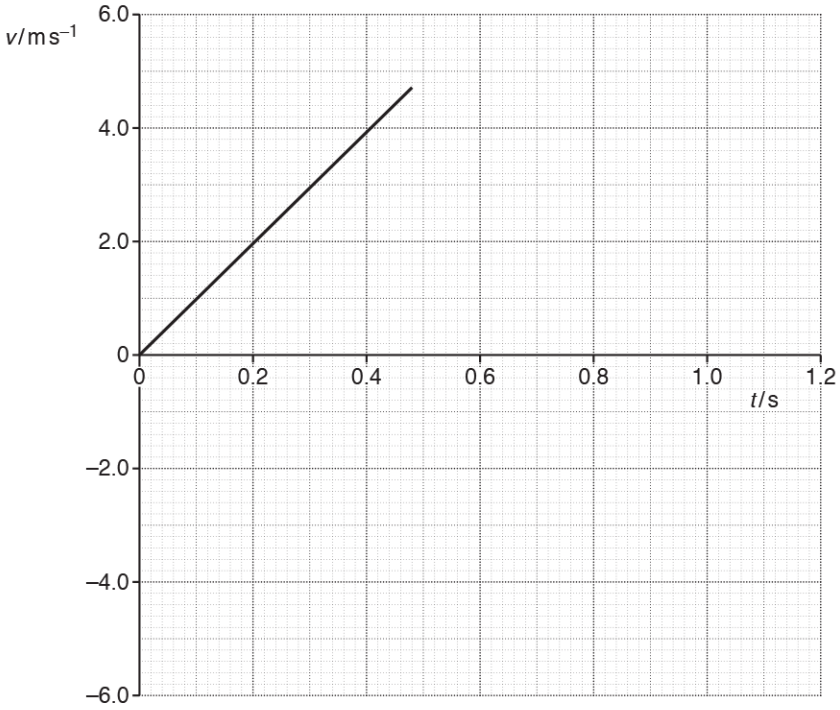

Fig. 1

Air resistance has a negligible effect on the motion of the tennis ball.

Use Fig. 1 to show that

i. the acceleration of the falling ball is about 10 m s^{-2}

[1]

ii. the kinetic energy of the ball just before impact with the surface is 0.63 J.

[2]

(b). The ball leaves the surface with 80% of the kinetic energy just before impact.

i. Calculate the magnitude of the velocity *v* of the ball as it leaves the surface.

v = ... m s^{-1} **[3]**

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ii. Complete Fig. 1 to show the variation of the velocity of the ball after it leaves the surface until it is at rest again.

[2]

iii. Determine the maximum height h reached by the ball after it bounces.

h **= ..... m [2]**

(c). The student repeats the experiment with a different ball that is affected by air resistance.

Explain how the graph in Fig. 1 now appears from the time the ball is released to the time it hits the surface.

.....

.....

.....

.....

..... **[2]**

10. In a hockey match a hockey ball is hit 18.0 m from the front of the goal. The ball leaves the hockey stick with initial velocity *v* at an angle θ to the horizontal ground. The ball passes over the goal at a maximum height of 2.0 m as shown in Fig. 3.

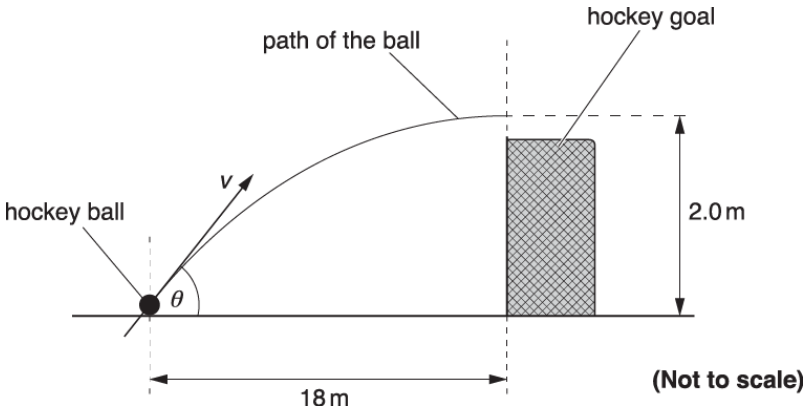

Fig. 3

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The hockey ball has a mass of 0.160 kg.

i. Calculate the initial kinetic energy E_k of the ball as it leaves the hockey stick.

E_k = ... J **[1]**

ii. Calculate the change in gravitational potential energy E_p of the ball as it moves from the ground to the maximum height.

E_p = ... J **[1]**

iii. Calculate the kinetic energy of the ball at the maximum height.

kinetic energy = ... J **[1]**

11(a). Fig. 3.1 shows a stunt rider on a powerful motorbike at **X** at the top of a ramp.

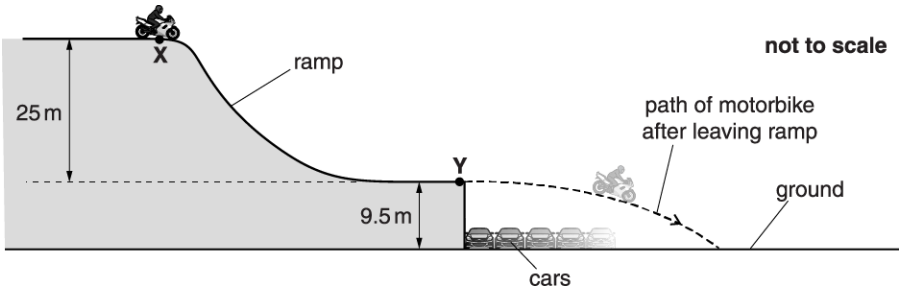

Fig. 3.1

The total mass of motorbike and rider is 190 kg. The height difference between the top and the bottom of the ramp is 25 m. The rider uses the engine to accelerate down the ramp. He leaves the end of the horizontal section of the ramp at **Y** with a speed of 30 m s^{-1}.

Calculate the loss of gravitational potential energy E_p of the rider and motorbike from **X** to **Y**.

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E_p = ... J **[1]**

(b). Calculate the kinetic energy E_k of the rider and motorbike at **Y**.

E_k = ... J **[1]**

(c). Explain why your answer to **(b)** is **greater** than your answer to **(a)**.

---**[1]**

(d). The total distance travelled by the motorbike from **X** to **Y** along the ramp is 120 m. Calculate the average accelerating force that the motorbike engine provides along the ramp.

force = ... N **[2]**

(e). The end of the ramp at **Y** is at a height of 9.5 m above the ground. The motorbike travels through the air after leaving the ramp and flies over a number of cars lined up side by side. Assume air resistance has negligible effect on the motion of the rider and motorbike.

i. Show that the time taken for the motorbike to travel from **Y** to the ground is 1.4 s.

[2]

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ii. Each of the cars has a width of 1.8 m. Estimate the number of cars the motorbike can clear.

number = ... **[2]**

12. **Fig. 22.2** shows an arrangement used to investigate how the kinetic energy of a toy car varies with its distance *d* from the top of the ramp.

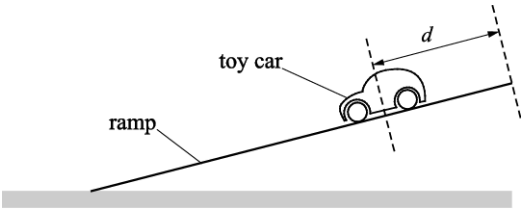

Fig. 22.2

The toy car is released from rest from the top of the ramp. The two graphs in **Fig. 22.3** show the variation of the gravitational potential energy E_p of the toy car and its kinetic energy E_k with distance *d* from the top of the ramp.

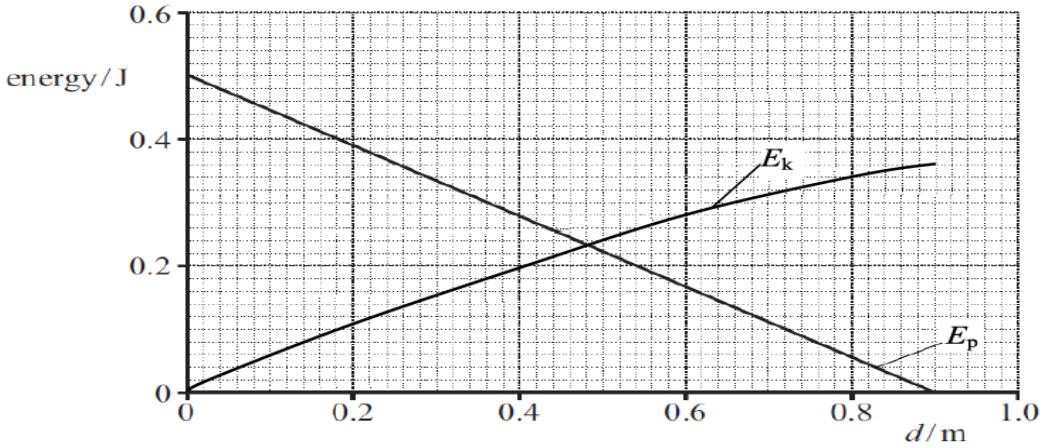

Fig. 22.3

The car travels a distance of 90 cm along the length of the ramp.

i. The variation of E_p with *d* is linear.
State why the E_k against *d* graph is **not** linear.

.....

..... **[1]**

ii. Use **Fig. 22.3** to determine the average resistive force acting on the toy car.

force = ...N **[2]**

END OF QUESTION PAPER

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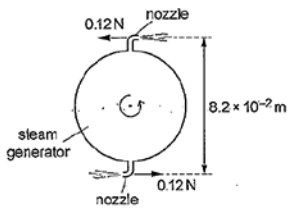
Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			**c**	1	
			Total	**1**	
2			**D**	**1**	
			Total	**1**	
3			**D**	1	**<u>Examiner's Comments</u>** This question was based on work done by a couple, and as such proved to be quite challenging. The work done by the couple is given by the expression below: work done = 2 × work done by each force = 2 × [0.12 × π × 8.2 × 10^{-2}] = 6.2 × 10^{-2} J The most popular answers turned out to be either A or C. The answer C was for the work done by one of the forces. This question was only accessible to the very top-end candidates. The exemplar 1 below shows an incorrect analysis that led to B being inserted into the answer box. **Exemplar 1** The diagram below shows a rotating steam generator.  The steam ejected from the nozzles provides a couple. The force at each nozzle is 0.12 N. The perpendicular distance between the nozzles is 8.2 × 10^{-2} m. What is the work done by the forces as the steam generator completes one revolution? **A** 0 J × **B** 9.8 × 10^{-3} J **C** 3.1 × 10^{-2} J **D** 6.2 × 10^{-2} J W = F × d 0.12 × 8.2 × 10^{-2} = 9.8 × 10^{-3} Your answer B [1] The candidate has either written the equation for work done, or torque of a couple. Substitution shows that the torque has been calculated. Unfortunately, the response of 9.8 × 10^{-3} J was there as one of the options. This exemplar shows that if the starting point is incorrect, it can easily lead to what looks like a promising response.
			Total	**1**	

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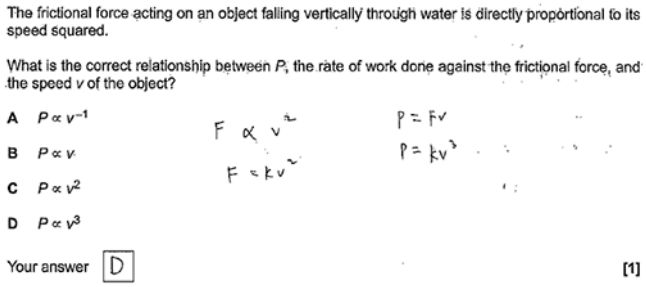
4			**A**	**1**	
			Total	**1**	
5			**D**	1	**<u>Examiner's Comments</u>** This question is based on the equation $P = Fv$, which also appears in the Data, Formulae and Relationship Booklet. In the question, information is given about the frictional force F, which is directly proportional to v^2. Therefore, the rate of work done P must be proportional to v^3; making D as the answer. Most candidates struggled with this question, with all the distractors being equally popular. Less than a quarter of the candidates, mainly from the upper quartile, scored a mark in this question. The exemplar 2 below the correct response from a candidate. **Exemplar 2**  The frictional force acting on an object falling vertically through water is directly proportional to its speed squared. What is the correct relationship between P, the rate of work done against the frictional force, and the speed v of the object? **A** $P \propto v^{-1}$ **B** $P \propto v$ **C** $P \propto v^2$ **D** $P \propto v^3$ $F \propto v^2$ $F = kv^2$ $P = Fv$ $P = kv^3$ Your answer D [1] This candidate demonstrates how this question can be tackled with minimal amount of work. The key equation is on the script, as is the relationship between F and v. The final answer appears in the box; a perfect technique.
			Total	**1**	
6			C	1	**Examiner's Comments** All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions. The correct key was **C** and the most popular distractor was **A**. The kinetic energy of the ball at the ground was K. At maximum height, the ball just has horizontal component of velocity. The kinetic energy of the ball is proportional to speed^2. At the maximum height the kinetic energy must therefore be $\cos^2 30° \, K = 0.75 \, K$.
			Total	**1**	

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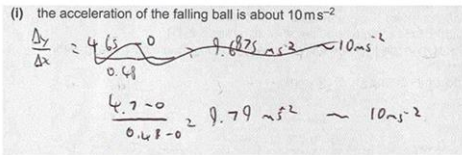
7			B	1	**Examiner's Comments** All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions.
			Total	**1**	
8			A	1	
			Total	**1**	
9	a	i	$\frac{\Delta v}{\Delta t}$ <u>and</u> $\Delta t \geq 0.20$ s 9.8 m s^{-2}	M1 A0	**Allow** tolerance of ±½ a small square e.g. $\frac{4.7(-0)}{0.48(-0)} = 9.79$ **<u>Examiner's Comments</u>** This question was a "show" type question. Candidates needed to show their working logically. Ideally candidates would state that that the acceleration was equal to the gradient, and then show the substitution of the data values for the gradient calculation. It was expected that candidates would have gained an answer of 9.79 m s^{-2} **Exemplar 1**  This candidate has clearly demonstrated from *Δy / Δx* that the gradient is to be determined. Co-ordinates are substituted into the gradient expression and it is clear that the candidate has used more than half the hypotenuse. The candidate then correctly evaluated the expression to give of 9.79 m s^{-2}. and then states that this is about 10 ms^{-2}. **AfL** Determining a gradient. Candidates should clearly demonstrate the co-ordinates that are used to calculate the gradient. The co-ordinates must lie on the line. A common error is when a candidate

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					uses a data point from a table of results. Candidates should be encouraged to read carefully the quantities from the axes and to pay attention to powers of ten and units. The length of the hypotenuse used for the gradient calculation should be at least half the length of the line. Candidates should clearly show the substitution of the co-ordinates and then evaluate the answer using the expression: gradient $= \frac{y_2 - y_1}{x_2 - x_1}$. The advantage of this method, it that negative gradients are automatically determined. **AfL** The gradient of a velocity-time graph is acceleration.
		ii	4.7 or $\frac{1}{2}$ x 0.057 x v^2 $\frac{1}{2}$ x 0.057 × 4.7^2 = 0.629565 0.63 J	M1 M1 A0	**<u>Examiner's Comments</u>** This was also a "show" type of question. Candidates needed to correctly read the maximum velocity (4.79 m s^{-1}) from the graph and change the mass of 57 g into kilograms. To gain the marks, clear substitution into the kinetic energy equation was needed with a correctly evaluated answer. **Exemplar 2** (ii) the kinetic energy of the ball just before impact with the surface is 0.63 J. $v = 4.7 ms^{-1}$ $KE = \frac{1}{2}(57\times10^{-3})(4.7)^2$ $KE = \frac{1}{2}mv^2$ $= 0.62957$ $\approx 0.63 J$ [2] In this two-mark answer, the candidate has clearly demonstrated the value from the graph as well as the equation that is going to be used. The candidate has correctly changed 57 g to kilograms effectively by using standard form. The candidate has then correctly evaluated the expression as 0.62957 before stating that this is approximately equal to 0.63 J. Candidates often find it helpful to underline relevant

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					quantities. In this response the candidate has underlined 0.63 J.
	b	i	0.8 x 0.63 J (0.504 J) OR $v^2 = \frac{2\times KE}{0.057}$ $v^2 = \frac{2\times 0.504}{0.057}$ 4.2(1) (ms^{-1})	C1 C1 A1	**Allow** one mark for correct rearrangement of KE equation with incorrect KE 17.684 **<u>Examiner's Comments</u>** In this question, higher ability candidates initially determined the kinetic energy (0.504 J) as the ball leaves the surface, before rearranging the kinetic energy equation. A few candidates did not take the final square root.
		ii	Straight line from (0.48, –4.2) to *x*-axis <u>and</u> plotted to ±½ small square *x*-axis intercept at *t* = 0.91 ± 0.03 (s) from negative *v*	C1 A1	**Allow ECF** from **(b)(i)** **Allow** (0.49, –4.2) / (0.50, –4.2) / (0.51, –4.2) / (0.52, –4.2) **Allow ECF** for incorrect negative *v* **<u>Examiner's Comments</u>** In this question, a large number of candidates did not understand that velocity is a vector quantity and drew a line with a negative gradient back towards the x-axis. The velocity of the ball as it leaves the surface is in the opposite direction and is therefore – 4.2 m s^{-1}. Candidates then needed to draw a parallel line to the initial line (since the acceleration is still the same). **AfL** Vector quantities have both a magnitude and a direction.
		iii	area under the graph = $\frac{1}{2}$ × 4.2 × 0.43 0.90 (m)	C1 A1	**Allow ECF** from **(i) and (ii)** **Allow** use of equation of motion: e.g. $s = \frac{4.2^2}{2\times 9.81}$ or *s* = (–4.2 × 0.43) + $\frac{1}{2}$ × 9.81 × 0.43^2 (numbers must be seen) **Allow** use of loss of KE = gain in PE **Allow** one significant figure **Note** 0.84 for Δ*t* = 0.40 to 0.97 for Δ*t* = 0.46 **<u>Examiner's Comments</u>**

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					There were many methods in which candidates could gain the marks in this question. It was helpful for clear methods to be demonstrated. The simplest was to determine the area under the velocity-time graph. Candidates also used the equations of uniform motion. Common errors seen included the incorrect velocity and when using the equations of motion but being confused about negative signs. Examiners on this occasion allowed an answer of 0.9 m which is one significant figure. Since the data used is to two significant figures, the final answer should also be to two significant figures. **AfL** The area under a velocity-time graph is displacement.
	c		Line will curve / be non-linear OR (magnitude of) gradient of line decreases (with increase in time) (Line will end with) a lower maximum/final velocity or hit the ground after a longer time	B1 B1	**Allow** sketch or gradient decreases / changes **Not** gradient is smaller / less steep / shallower / lower **Allow** ball will have a lower maximum/final velocity or hit the ground after a longer time) **<u>Examiner's Comments</u>** Candidates found this question challenging. Many candidates answered the question in terms of air resistance and terminal velocity. The question required candidates to explain how the graph would appear. Several candidates stated that the gradient would be smaller but did not clearly state that the gradient would decrease over time and not indicate that the line would curve. Candidates needed to also indicate that the line would indicate a lower maximum velocity at a longer time.
			Total	**12**	
10		i	$(E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.16 \times 30.7^2)$ E_k = 75 (J)	B1	Possible ECF from **(iii)** **Examiner's Comments** This was generally well answered.
		ii	($E_p = mgh$ = 0.16 × 9.81 × 2.0 =) 3.1 (J)	B1	**Allow** $(E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.16 \times 6.3^2) = 3.2(\text{J})$

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					Examiner's Comments Most candidates correctly calculated the gravitational potential energy although some weaker candidates used the answer from (i).
		iii	**(i) − (ii)**; (75 − 3.1) or $(E_k = \frac{1}{2} \times 0.16 \times 30^2)$ kinetic energy = 72 (J)	B1	Possible ECF from **(i)** and **(ii)** **Note:** Answer is 63 (J) when 28 (m s^{-1}) is used from **(ii)** **Examiner's Comments** Candidates either used the kinetic energy equation of subtracted the change in gravitational potential energy form their answer in (i). Common errors were either to state that the kinetic energy was zero or equal to the change in potential energy calculated in (ii).
			Total	**3**	
11	a		E_p = 190 × 9.81 × 25 E_p = 4.7 × 10^4 (J)	B1	**Note**: Answer is 4.66 × 10^4 to 3sf **Examiner's Comments** All the candidates knew which formula to use but some used the incorrect height. Others arrived at the correct answer but wasted time calculating the change in gravitational potential energy (GPE) over 34.5 m, then subtracting the change in GPE over 9.5 m.
	b		E_k = ½ × 190 × 30^2 E_k = 8.6 × 10^4 (J)	B1	**Note:** Answer is 8.55 **x** 10^4 to 3sf **Examiner's Comments** Almost universally correct answers.
	c		Work done by the motorbike / energy from the engine (AW)	B1	**Note**: There must be reference to work or energy **Allow:** chemical energy to kinetic energy / E_K **Examiner's Comments** A more testing question. Candidates attempted to use arguments about the motorcycle accelerating but failed to mention energy or work so didn't score the mark. Many candidates explained the difference in energy by discussing heat losses - which would have resulted in less kinetic energy rather than more. This question discriminated well between the upper, mid and lower quartiles.
	d		work done = change in energy force × 120 = (8.55 - 4.66) × 10^4	C1	Possible ecf from **(a)** and **(b)**

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			force = 320 (N)	A1	**Examiner's Comments** Another excellent discriminating question. Many candidates wrongly applied equations of motion to the situation, using the length of the ramp as displacement in an attempt to determine the acceleration and hence the force, resulting in answers around 712 N.
	e	i	(s = ½ at^2 – for the vertical fall)		
		i	9.5 = ½ × 9.81 × t^2 (Any subject)	M1	
		i	$t = \sqrt{(2\times 9.5)/9.81}$ or 1.39	M1	
		i	time = 1.4 (s)	A0	**Examiner's Comments** A significant number of candidates used 30 m s^{-1} as the initial velocity leading to an incorrect answer. Candidates who did not rearrange the formula to make t the subject of the equation did not score the second mark.
		ii	Horizontal velocity = 30 m s^{-1}		
		ii	distance = 1.4x30 or 42 (m)	C1	
		ii	(number of cars =) 42/1.8		
		ii	(Number of car =)42/1.8	A1	**Allow**: 23.3 cars **Allow:** 22 if height of last car is mentioned **Examiner's Comments** There were many good answers but a number of candidates who had failed to determine 1.39 s in the previous question omitted this question.
			Total	**9**	
12		i	There is friction. GPE is transferred to KE and heat or thermal (energy).	B1	
		ii	work done = (0.50 − 0.36) (J) or work done = 0.14 (J)	C1	
		ii	F × 0.90 = 0.14, therefore resistive force = 0.16 (N)	A1	
			Total	**3**	

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